

# Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)



## Compound angle formula for cosine

- 1 Use the compound angle formula  $\cos(A + B) = \cos A \cos B - \sin A \sin B$  to find the exact value of  $\cos 75^\circ$ , writing  $75^\circ = 45^\circ + 30^\circ$ .
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## Compound angle formula for sine

- 2 Use the compound angle formula  $\sin(A - B) = \sin A \cos B - \cos A \sin B$  to find the exact value of  $\sin 15^\circ$ , writing  $15^\circ = 45^\circ - 30^\circ$ .
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## Compound angle formula for tangent

- 3 The compound angle formula for tangent is  $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$ . Use it with  $A = 45^\circ$  and  $B = 30^\circ$  to find the exact value of  $\tan 15^\circ$ , giving your answer in simplified surd form.
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## Proving identities using the compound angle formulas

- 4 Prove that  $\sin(A + B) + \sin(A - B) = 2\sin A \cos B$ .
- 5 Prove that  $\cos(A + B)\cos(A - B) = \cos^2 A - \sin^2 B$ .
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## Solving a trigonometric equation using a compound angle formula

- 6 Solve  $\sin(x + 30^\circ) = \cos x$  for  $0^\circ \leq x \leq 360^\circ$ .
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## Evaluating reciprocal trigonometric functions from a given ratio

- 7 Given  $\cos \theta = \frac{3}{5}$  with  $\theta$  in Quadrant I, find the exact values of  $\sec \theta$ ,  $\csc \theta$ , and  $\cot \theta$ .
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## Reciprocal trigonometric functions in another quadrant

- 8 Given  $\tan \theta = -2$  with  $\theta$  in Quadrant II, find the exact values of  $\sec \theta$ ,  $\csc \theta$ , and  $\cot \theta$ .

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### Deriving $1 + \tan^2\theta = \sec^2\theta$

- 9 Starting from the Pythagorean identity  $\sin^2\theta + \cos^2\theta = 1$ , divide through by  $\cos^2\theta$  to derive  $1 + \tan^2\theta = \sec^2\theta$ .

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### Deriving $1 + \cot^2\theta = \csc^2\theta$

- 10 Starting from the Pythagorean identity  $\sin^2\theta + \cos^2\theta = 1$ , divide through by  $\sin^2\theta$  to derive  $1 + \cot^2\theta = \csc^2\theta$ .

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### Simplifying an expression using reciprocal identities

- 11 Simplify  $\sec\theta - \cos\theta$ , expressing your answer in terms of  $\sin\theta$  and  $\tan\theta$ .

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### Mixed application: combining reciprocal identities across quadrants

- 12 Given  $\sec\theta = 13/5$  with  $\theta$  in Quadrant IV, find the exact value of  $\sin\theta + \cot\theta$ .

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### Compound angle formula for sine (sum form)

- 13 Use the compound angle formula  $\sin(A + B) = \sin A \cos B + \cos A \sin B$  to find the exact value of  $\sin 75^\circ$ , writing  $75^\circ = 45^\circ + 30^\circ$ .

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### Compound angle formula for tangent (sum form)

- 14 Use  $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$  with  $A = 45^\circ$  and  $B = 30^\circ$  to find the exact value of  $\tan 75^\circ$ , giving your answer in simplified surd form.

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### Finding an exact value in Quadrant II

- 15 Use the compound angle formula  $\cos(A + B) = \cos A \cos B - \sin A \sin B$  to find the exact value of  $\cos 105^\circ$ , writing  $105^\circ = 60^\circ + 45^\circ$ .

## Deriving the double-angle formula from the compound angle formula

- 16 By setting  $A = B = \theta$  in  $\sin(A + B) = \sin A \cos B + \cos A \sin B$ , derive the double-angle formula  $\sin 2\theta = 2\sin \theta \cos \theta$ .
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## A second identity from the compound angle formulas

- 17 Prove that  $\cos(A - B) - \cos(A + B) = 2\sin A \sin B$ .
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## Solving another trigonometric equation using a compound angle formula

- 18 Solve  $\sin(x + 60^\circ) = \cos x$  for  $0^\circ \leq x \leq 360^\circ$ .
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## A quick simplification from the Pythagorean identity

- 19 Simplify  $(\sec \theta + \tan \theta)(\sec \theta - \tan \theta)$ .
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## Reciprocal trigonometric functions in Quadrant III

- 20 Given  $\sec \theta = -13/12$  with  $\theta$  in Quadrant III, find the exact values of  $\sin \theta$ ,  $\cos \theta$ , and  $\tan \theta$ .
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## Connecting a co-function identity to the reciprocal functions

- 21 Using the compound angle formula for cosine, show that  $\cos(90^\circ - \theta) = \sin \theta$ , and hence prove that  $\sec(90^\circ - \theta) = \csc \theta$ .
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## Solving an equation involving a reciprocal function

- 22 Solve  $\csc \theta = 2$  for  $0^\circ \leq \theta \leq 360^\circ$ .
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## Proving a reciprocal identity from scratch

- 23 Prove that  $\tan \theta + \cot \theta = \sec \theta \csc \theta$ .
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### A capstone exact-value application

- 24 Find the exact value of  $\cos 15^\circ$  (writing  $15^\circ = 45^\circ - 30^\circ$ ), then use it together with  $\cos 75^\circ = (\sqrt{6} - \sqrt{2})/4$  found earlier to compute  $\cos 75^\circ - \cos 15^\circ$  exactly.